Lab 8 — Student Submission Template

Group no.: 1 Project title: Abhinay – Film Production Networking and Casting Platform

Group members and roles: Ishan – Project Manager + Backend/Architecture + Integration; Happy – Frontend (Authentication & Casting); Abhishek – Frontend (Messaging & Admin); Arjun – Backend (Authentication & Profiles); Triya – Frontend (Casting & Applications) + Backend (Media & Audition); Priyadarshani – Backend (Projects & Networking); Himanshu – Backend (Media & Audition); Kshitesh – Full-stack (Projects & Networking); Yuva Raju – Backend (Applications & Notifications), also Estimation & Cost Manager; Nikitha – Frontend + QA (Integration, Testing & UI).

Important: Continue with your ORIGINAL group project. Use the WBS and schedule prepared in earlier labs. Fill in realistic estimates and write down any assumptions.

A. Resource requirement list

Human resources = people. Non-human resources = devices, software, computers, servers, rooms, etc.

No.	Resource / person	Type	Purpose / skill needed
1	Ishan	Human	Project manager; backend/architecture and integration of modules; real-time messaging backend.
2	Backend developers: Arjun (auth & profiles), Priyadarshani (projects & networking), Himanshu and Triya (media & audition), Yuva Raju (applications & notifications)	Human	Node.js/Express APIs, database access, JWT/RBAC, media-storage integration, notifications.
3	Frontend developers: Happy (auth & casting), Triya (casting & applications), Abhishek (messaging & admin)	Human	React UI: posting forms, search, application screens, inbox; responsive design.
4	Kshitesh (full-stack, projects & networking)	Human	Project page and connect/follow features, front end and back end.
5	Nikitha (frontend + QA)	Human	Integration and UI work, test cases, bug reports, feed UI.
6	Laptops / PCs (one per member)	Non-human (hardware)	Development, local testing and documentation.
7	VS Code, Git and GitHub	Non-human (software)	Coding, version control, code review and branching.
8	Node.js/Express, React, Prisma and database	Non-human (software)	Application stack: backend, frontend and

			database access.
9	Shared dev/staging server	Non-human (server)	Hosting integrated builds for testing; one shared instance for the group.
10	Cloud media-storage service and real-time messaging layer (e.g., Socket.IO)	Non-human (external service)	Storing/streaming photos, reels and self-tapes; live chat and audition communication.
11	Postman, Figma, browsers and test phone/device; internet	Non-human (software/hardware)	API testing, UI design, responsive/mobile testing, online access to all tools.

B. Activity and resource allocation

Choose 5–8 activities from your own project WBS. Assign one main owner, planned week and estimated effort in hours.

WBS task / activity	Assigned member	Planned week	Hours	Tool / device
A. Requirement Analysis & Project Planning	Ishan – 6 h/wk (planning, coordination) Yuva Raju – 4 h/wk (estimation & cost) Nikitha – 4 h/wk (wireframes, test plan) Happy, Abhishek, Arjun, Triya, Priyadarshani, Himanshu, Kshitesh – 3 h/wk each (requirements, design review)	Weeks 1–3	105 total (35 h/wk)	Laptop, documentation tools, Figma
B. 1.1 User Management & Profiles	Arjun – 9 h/wk (backend auth & profiles) Happy – 8 h/wk (authentication UI) Ishan – 3 h/wk (architecture review)	Weeks 4–5	40 total (20 h/wk)	VS Code, Node/Express, React, database, Postman
C. 1.2 Casting Marketplace	Week 6 (similar in Week 7): Happy – 8 h (1.2.1.1 posting UI 6 + 1.2.1.2 edit/close UI 2) Triya – 7 h (1.2.2.1 browse/search UI 4 + 1.2.2.2 apply UI 3) Yuva Raju – 7 h (1.2.2.2 application API 4 + 1.2.3.1 shortlist folders 3) Arjun – 3 h (RBAC checks on new endpoints) Ishan – 4 h (project monitoring and integration)	Weeks 6–7	58 total (29 h/wk)	VS Code, React, Node/Express, database, GitHub
E. 1.4 Communication & Audition	Week 6 (similar in Week 7): Ishan – 8 h (1.4.1.1 send/receive messages, backend/real-time) Abhishek – 6 h (1.4.1.2 message inbox UI) Himanshu – 13 h (1.4.3.1 self-tape upload 5 + 1.4.3.2 secure media storage 3 + 1.4.2.1 schedule audition session 5)	Weeks 6–7	54 total (27 h/wk)	VS Code, Node/Express, React, real-time messaging layer, cloud media storage, staging server
F. 1.5 Project Pages & Networking	Priyadarshani – 12 h (1.5.1.1 project page API 4 + 1.5.2.1 link crew profiles 4 + 1.5.3.1 connect/follow API 4) Kshitesh – 8 h (1.5.1.1 project page UI 5 +	Week 6	31 total	VS Code, React, Node/Express, database, staging

	1.5.3.1 connect/follow UI 3) Nikitha – 9 h (1.5.3.2 feed UI 4 + testing of Week 6 features 5) Arjun – 2 h (profile API for crew linking)			server, test phone
D. 1.3 Application Lifecycle Management	Yuva Raju – 9 h/wk (status tracking and notifications backend) Triya – 8 h/wk (applicant review dashboard UI) Arjun – 4 h/wk (backend support) Ishan – 3 h/wk (architecture/integration)	Weeks 8–9	48 total (24 h/wk)	VS Code, GitHub, database, Postman
G. 1.6 Admin & Moderation	Abhishek – 9 h/wk (admin frontend) Arjun – 6 h/wk (moderation and credit-verification APIs) Ishan – 3 h/wk (architecture support) Nikitha – 3 h/wk (UI and QA)	Weeks 10–11	42 total (21 h/wk)	VS Code, React, Node/Express, database
H. Integration, Testing, Deployment & Final Documentation	Nikitha – 9 h/wk (QA and testing) Ishan – 8 h/wk (integration and deployment) All other 8 members – 3 h/wk each (bug fixing, final documentation)	Weeks 12–14	123 total (41 h/wk)	GitHub, browsers, test phone, staging server

C. Workload check

Choose one planned week: Week 6. Each member has 10 hours available in that week. Reason: in the Lab 7 CPM schedule, C (1.2 Casting Marketplace), E (1.4 Communication & Audition) and F (1.5 Project Pages & Networking) all start together at day 25, so Week 6 is the busiest week. Each member's assigned hours below are the Week 6 hours added up from the C, E and F rows of Table B.

Member	Total assigned hours	Available hours	Difference (assigned − available)	Overloaded?
Ishan	12	10	12 − 10 = +2	Yes – 2 h over (120%)
Happy	8	10	8 − 10 = −2	No (80%)
Abhishek	6	10	6 − 10 = −4	No (60%)
Arjun	5	10	5 − 10 = −5	No (50%)
Triya	7	10	7 − 10 = −3	No (70%)
Priyadarshani	12	10	12 − 10 = +2	Yes – 2 h over (120%)
Himanshu	13	10	13 − 10 = +3	Yes – 3 h over (130%)
Kshitesh	8	10	8 − 10 = −2	No (80%)
Yuva Raju	7	10	7 − 10 = −3	No (70%)
Nikitha	9	10	9 − 10 = −1	No (90%)

Example: 14 hours assigned and 10 hours available means 4 hours overloaded.

D. Check shared tools and overlapping work

Person or shared tool	Conflicting tasks / times	What is the problem?
Himanshu + cloud media storage	1.4.3.1 Self-tape upload, 1.4.3.2 Secure media storage and 1.4.2.1 Schedule audition, all in Week 6.	One person has 13 h of work against 10 h available, and the upload and storage tasks use the same media-storage account, so they cannot run in parallel.
Shared dev/staging server (and database)	Ishan, Himanshu, Priyadarshani, Kshitesh, Yuva Raju, Abhishek and Nikitha all deploy to or test on the single staging server in Week 6.	Simultaneous deployments can overwrite each other's builds and test data, and break each other's tests.
Dependent tasks (several members)	1.4.1.2 Inbox UI (Abhishek) needs the messaging API (Ishan); 1.5.1.1 Project page UI (Kshitesh) needs the project API (Priyadarshani); 1.5.2.1 Crew linking needs Arjun's profile API; Apply UI (Triya) needs Yuva Raju's application API.	A dependent task cannot start at the same time as its predecessor; planning them in parallel gives idle time or rework.
Test phone and integration (Nikitha, Ishan)	Nikitha's testing and Ishan's integration both need a stable staging build and the single test phone.	Testing mid-week would hit unfinished features, and two people cannot use the single test phone at once.

E. Balance the work

If someone is overloaded, move tasks to another student, share tasks, or shift dates (without quietly changing project scope).

Problem found	Change made	Reason for change
Ishan overloaded by 2 h (12 h vs 10 h).	Move 2 h of message API access control and validation (part of 1.4.1.1) to Arjun. Ishan keeps 6 h of the messaging backend.	Arjun is the backend owner for authentication and profiles, so access control is his skill, and he had 5 h free.
Priyadarshani overloaded by 2 h (12 h vs 10 h).	Move 2 h of 1.5.3.1 Connect/follow API to Kshitesh (full-stack, Projects & Networking). Priyadarshani keeps 2 h of it.	Kshitesh works on the same module and had 2 h free, so there is no learning gap.
Himanshu overloaded by 3 h (13 h vs 10 h).	Move 3 h of 1.4.2.1 Schedule audition (scheduling logic and notifications) to Triya. Himanshu keeps 2 h of it.	Triya is also a backend developer for Media & Audition and had 3 h free. Scope is unchanged.
Shared staging server, test phone and dependencies.	Time slots: deploy to staging Mon–Tue (Arjun, Yuva Raju, Priyadarshani, Ishan, Himanshu), Wed–Thu (Happy, Triya, Abhishek, Kshitesh); Fri: Ishan integration, then Nikitha testing on the test phone. Dependent UI tasks start after the API is ready, using mock data until then.	Prevents builds from overwriting each other, avoids waiting on unfinished work and gives testing a stable build. No dates in the Lab 7 critical path are moved.

F. Final allocation after changes

Student	Updated task(s)	Final assigned hours	Available hours	Within limit?
Ishan	1.0 Monitoring and integration (4 h) + 1.4.1.1 Messaging backend (6 h)	10	10	Yes (100%)
Happy	1.2.1.1 and 1.2.1.2 Posting UI (no change)	8	10	Yes (80%)
Abhishek	1.4.1.2 Message inbox UI (no change)	6	10	Yes (60%)
Arjun	RBAC (3 h) + profile API (2 h) + messaging access control from 1.4.1.1 (2 h)	7	10	Yes (70%)
Triya	1.2.2.1 and 1.2.2.2 UI (7 h) + 1.4.2.1 scheduling logic and notifications (3 h)	10	10	Yes (100%)
Priyadarshani	1.5.1.1 Project API (4 h) + 1.5.2.1 Crew linking (4 h) + 1.5.3.1 Connect/follow API (2 h)	10	10	Yes (100%)
Himanshu	1.4.3.1 Self-tape (5 h) + 1.4.3.2 Media storage (3 h) + 1.4.2.1 Audition session (2 h)	10	10	Yes (100%)
Kshitesh	1.5.1.1 Project page UI (5 h) + 1.5.3.1 Connect/follow UI (3 h) + Connect/follow API (2 h)	10	10	Yes (100%)
Yuva Raju	1.2.2.2 Application API + 1.2.3.1 Shortlist folders (no change)	7	10	Yes (70%)
Nikitha	1.5.3.2 Feed UI (4 h) + testing (5 h) (no change)	9	10	Yes (90%)

G. Assumptions and short conclusion

Assumptions (example: each student can give 10 hours per week)

1. Each member can give 10 hours per week to the project; Week 6 is analysed because C, E and F run in parallel in the Lab 7 schedule (1 week = 5 working days).

2. Table B covers the whole 14-week plan (Lab 7: 1 week = 5 working days). Hours are planning estimates based on the Lab 4 size estimate (4,750 LOC) and the team's experience. For two-week modules, Week 7 is assumed similar to Week 6, and the balancing done for Week 6 is applied in Week 7 as well.

3. All 10 group members are included in the workload check, using the group's responsibility table (for example, Triya is also a backend developer for Media & Audition).

4. Tasks are moved only to members whose role fits (for example, access control to the Authentication & Profiles backend owner), so no learning gap is added.

5. Only one shared staging server, one media-storage account and one test device are available; tasks are not moved outside the approved scope.

Project manager: In 2–3 sentences, explain what you learned and how this plan will help your project.

We learned that Week 6 was overloaded for Ishan (12 h), Priyadarshani (12 h) and Himanshu (13 h), while Arjun, Triya, Yuva Raju and Abhishek had 3–5 h free, even though the total (87 h) fits the group's capacity (100 h).

Moving 2–3 h of suitable work to Arjun, Kshitesh and Triya, and giving the shared staging server and test phone time slots, brought every member within 10 h (60–100% utilization) without changing the project scope.

This plan helps the project by keeping the critical path (A–B–C–D–G–H) on time, since C and E in Weeks 6–7 are not delayed by an overloaded person or a blocked server.

Submission checklist

☒ Resource list is complete

☒ Every main WBS activity has an owner

☒ Task hours and workload totals are consistent

☒ Any overload or resource conflict has been checked

☒ Changes and assumptions are explained

☒ One document is submitted per group